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The Double-helical Nature of the Crystalline Part of A-Starch

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A new three-dimensional structure of the crystalline part of A-starch is described in which the unit cell contains 12 glucose residues located in two left-handed, parallel-stranded double helices packed in a parallel fashion; four water molecules are located between these helices. Chains are crystallized in a monoclinic lattice with $a = 2{\cdot}124$ nm, $b = 1{\cdot}172$ nm, $c = 1{\cdot}069$ nm and $\gamma = 123{\cdot}5°$, the c axis being parallel to the helix axis. Systematic absences are consistent with the space group $B2$. The structure was derived from joint use of electron diffraction of single crystals, X-ray powder patterns decomposed into individual peaks and previously reported X-ray fibre diffraction data after adequate re-indexing. The repeating unit consists of a maltotriose moiety where the glucose residues have the $^4C_1$ pyranose conformation and are $\alpha(1 \rightarrow 4)$ linked. The conformation of the glycosidic linkage is characterized by torsion angles (Φ, Ψ) which take the values ($91{\cdot}8, -153{\cdot}2$), ($85{\cdot}7, -145{\cdot}3$) and ($91{\cdot}8, -151{\cdot}3$); all the primary hydroxyl groups exist in a *gauche–gauche* conformation. There are no intramolecular hydrogen bonds. Within the double helix, interstrand stabilization is achieved without any steric conflict and through the occurrence of $O_{(2)} \cdots O_{(6)}$ type hydrogen bonds. The present structure is consistent with both physicochemical and biochemical aspects of the crystalline component of the cereal starch granules.

1. Introduction

Starch granules are the major form for packaging storage carbohydrates in green plants. The structure of starch has been the subject of many investigations and much speculation. A well-documented review on the present knowledge of the organization of the granule has appeared (French, 1984). Most granules exhibit a natural crystallinity with different diffraction patterns, according to their botanical origin: A-type from cereals and B-type from tubers. Less frequently, an intermediate polymorph, C-starch, is found in some plants. In all instances, starch has two major carbohydrates components: amylose and amylopectin. Amylose is a long, linear homopolymer of $(1 \rightarrow 4)$ linked α-D-glucopyranosyl units. Amylopectin exhibits a branched "tree-like" structure of linear sequences with the same $(1 \rightarrow 4)$ linkages, connected together through $\alpha(1 \rightarrow 6)$ linkages. Amylopectin typically shows a bimodal distribution of chain lengths centred at 15 and 45 glucopyranosyl units (Lineback, 1986). Their counterpart in the animal kingdom, glycogen, displays the same basic structural features as amylopectin, but has a symmetrical distribution of unit chains of average length 10 to 14 glucose residues (Gunja-Smith *et al.*, 1971) which may be responsible for the absence of long-range ordering in the solid-state associated with the lack of crystallinity.

Despite the fact that the first reported X-ray diffraction diagrams of starch were obtained in the

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1930's (Katz & van Italie, 1930), attempts to gain an unambiguous understanding of the crystalline arrangement have not yet been completely successful. Several structural models have been postulated; Kainuma & French (1972) were the first to suggest a double-stranded helix for the B-polymorph. The presently accepted models for A and B starches were established on the basis of X-ray fibre diffraction data and extensive computerized molecular modelling (Wu & Sarko, 1978*a*,*b*). Both A and B structures were interpreted on the basis of right-handed parallel-stranded double helices, packed in an antiparallel fashion. While being nearly identical in molecular conformations, A and B amyloses differ considerably in the crystalline packing of the duplex. These models have been questioned by several authors (French, 1984; Wild & Blanshard, 1986).

From biosynthetic considerations, it would be more reasonable to expect parallel packing of parallel-stranded double helices, so that all the amylosic chains would have the same polarity. Similar arguments about chain polarity can be derived from biochemical data, which in current models rely only on parallel arrangements of molecules (Duprat *et al.*, 1980; Manners & Matheson, 1981).

Ongoing work on *in-vitro* growth of single crystals of amylose and subsequent diffraction studies (Buléon *et al.*, 1984; Tran & Buléon, 1987), along with recent crystallographic data on amylosic oligomers (Pangborn *et al.*, 1985), lead us to reinvestigate the structure of A-starch. Following a preliminary report (Imberty *et al.*, 1987) the present work describes in detail a new three-dimensional structure for A-starch which differs markedly from the currently accepted model (Wu & Sarko, 1978*b*).

2. Materials and Methods

(a) *Sample preparation*

Amylose fractions with degree of polymerization (*DP*) 15 were obtained by mild hydrolysis of potato starch as described by Duprat *et al.* (1980). A suspension of 1% (w/v) of this sample was dispersed in water and boiled for 10 min, after which the amylose was dissolved. The solution was hot-filtered and cooled to 50°C. A-amylose was then precipitated as a fine crystalline powder by adding an equal volume of hot methanol (or acetone). After 3 h at 50°C, the crystallization was complete. The precipitate, which had flocculated, was then collected and washed by successive centrifugations by mixture (1 : 1, v/v) of water and methanol (or acetone). The crystalline powder was then filtered and stored in a vessel maintained at a relative humidity of 90%.

For preparing single crystals of A-amylose, the crystallization recipe was similar except that the starting solution was more dilute (0·05%, w/v) and the crystallization was induced by a slow diffusion of acetone vapours at 60°C, while maintaining gentle stirring. A suspension of single crystals of A-amylose resulted within 12 h. They were washed by successive centrifugations with mixtures of 1 : 1 (v/v) acetone/water. The crystals were then stored as a suspension in such a mixture.

For density measurements, a batch of crystals was filtered and equilibrated in a moist atmosphere until its moisture content was of 24%. The density of the crystals was then measured by flotation in a mixture of methylene chloride and carbon tetrachloride.

(b) *X-ray diffraction*

X-ray powder diffraction diagrams were recorded on a CGR Sigma 2080 diffractometer. The $CuK\alpha_1$ radiation ($\lambda = 0{\cdot}15405$ nm) was selected with a monochromator and detected by a scintillation counter. After Lorentz and polarization corrections, the part of the diagram ranging from 5° to 25° of Bragg angles was refined with a new decomposition method (Tran & Buléon, 1987). The calculation of the position and the intensity of each individual reflection (with a Pearson VII type function for peak shape) was performed simultaneously with a unit cell refinement by using an optimization method which reduces the false minima risks. This program was specially written for diffraction diagrams of poor resolution, as often found with polymers and biopolymers.

(c) *Electron diffraction*

Drops of the crystalline suspension were deposited on carbon-coated grids. The grids were allowed to dry by equilibration in a desiccator maintained at 97% relative humidity. They were then observed with a Philips EM400T electron microscope, equipped with a cold stage tilting sample holder. In all cases, the specimen was observed in the frozen hydrated state under low dose conditions (Chanzy *et al.*, 1977). Images and electron diffraction diagrams were recorded at 120 kV on Ilfoset FT35 films. Electron diffractograms of tilted crystals were recorded by rotating the microscope goniometer with angles ranging from −60° to +60°. Meaningful tilted diagrams were obtained on crystals that were aligned with their long axis parallel to the cold stage holder axis.

(d) *Nomenclature*

A fragment of amylose chain (maltotriose) is represented in Fig. 1 with the labelling of the atoms and the torsional angles of interest. The relative orientation of 2 contiguous (1 → 4) linked α-D-glucose residues is given by torsional angles Φ and Ψ which are defined by the 4 atoms sequences $O_{(5)}$–$C_{(1)}$–$O_{(1)}$–$C_{(4')}$ and $C_{(1)}$–$O_{(1)}$–$C_{(4')}$–$C_{(5')}$, respectively. Other conformational parameters are the orientations of the primary hydroxyl groups around $C_{(5)}$–$C_{(6)}$ bonds. The conformation is referred to as either *gauche–trans*, *gauche–gauche* or *trans–gauche*. In this terminology, the torsion angle $O_{(5)}$–$C_{(5)}$–$C_{(6)}$–$O_{(6)}$ is stated first, followed by the torsion angle $C_{(4)}$–$C_{(5)}$–$C_{(6)}$–$O_{(6)}$ (Marchessault & Pérez, 1979). Furthermore, for hexopyranoses having a *gluco* configuration at $C_{(4)}$, the occurrence of a *trans–gauche* conformation at $C_{(6)}$ being very unlikely, it will not be envisaged throughout this work. The sign of the torsion angles is defined in agreement with the rules recommended by the *IUPAC-IUB Commission of Biochemical Nomenclature* (1970, 1971).

(e) *Molecular modelling*

Starting geometry for the glucose residue in its $^4C_1$ chair conformation was taken from the crystal structure of α-D-Glucose monohydrate reported by Hough *et al.*

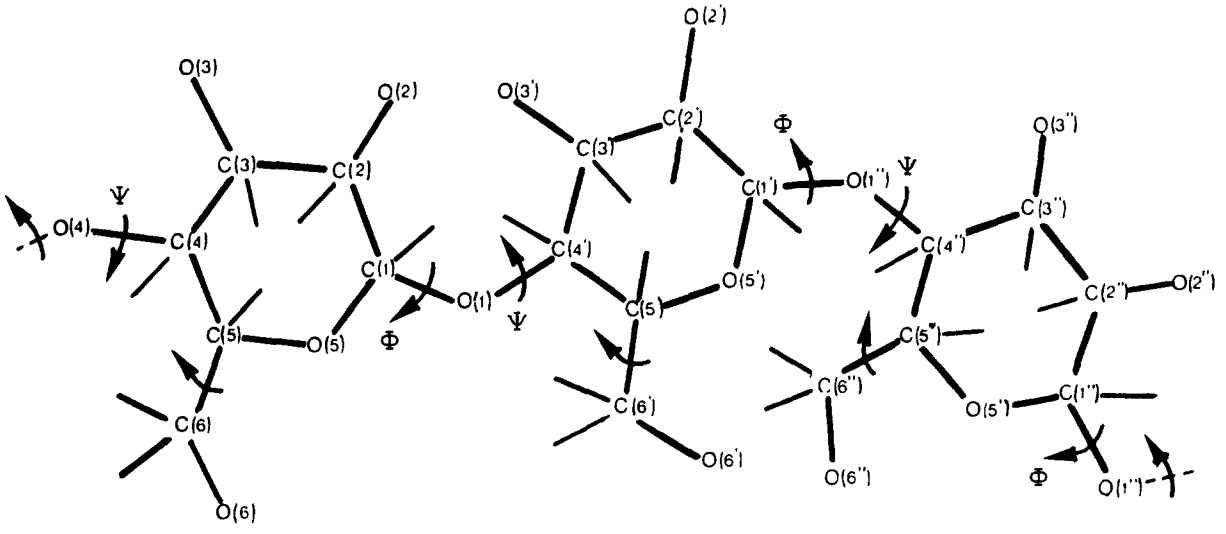


Figure 1. Schematic representation of the maltotrioside unit along with the labelling of the atoms and the torsional angles of interest. The hydrogen atoms are not labelled in this Figure.

(1973). The co-ordinates of the hydrogen atoms were determined using a C–H bond length of 0·10 nm and a bond vector related appropriately to the C–C and C–O bond vectors. Hydroxylic hydrogen atoms were not considered. The potential energy was calculated by including the partitioned contributions arising from the van der Waals' torsional and hydrogen contributions. The van der Waals' interactions were evaluated by using 6–12 potential functions with the parameters proposed by Scott & Scheraga (1966). A 3-fold sinusoidal potential was used for rotation about the glycosidic torsion angle Ψ, with a barrier of 1·0 kcal/mol (1 cal = 4·184 J) for the eclipsed conformations. For rotation about the glycosidic linkage $C_{(1)}$–$O_{(1)}$, i.e. Φ angle, the intramolecular mechanism responsible for the exo-anomeric effect was taken into account using the potential function proposed by Tvaroska (1984); a sinusoidal 3-fold eclipsed rotational barrier of 1·0 kcal/mol was also included. The hydrogen bond energy was computed by an empirical expression proposed by Pérez & Vergelati (1987), based on the oxygen atom distances. No electrostatic interaction was taken into account. For a disaccharide, the energy map was calculated as a function of Φ and Ψ at intervals of 5 deg. Iso-energy contours were drawn by interpolation at 1 kcal/mol with respect to the energy minimum.

Helical arrangements are described in terms of a set of helical parameters (n, h); n is the number of residues (i.e. backbone glucose units) per turn of the helix, and h is the translation along the helix axis. The chirality of the helix is defined by the sign of h. Arbitrarily, a right-handed helix will correspond to positive values of h; conversely, negative values of h will designate left-handed helices. Whenever the values $h = 0$ or $n = 2$ are intercrossed, the screw sense of the helix changes to the opposite sign. The helical parameters are calculated following an algorithmn reported by Gagnaire *et al.* (1980).

(f) *Refinement*

The packing of the chains was described by the variables (μ) for orientation about, and (w) for translation along, the helix axis. In addition, the position of water molecules was described by 3 parameters. The linked atom least-squares (LALS) optimization (Arnott & Wonacott, 1966; Smith & Arnott, 1978) was carried out by minimizing the discrepancy function Ω, where Ω is given by:

$$\Omega = \Sigma W_m(|{}_oF_m|^2 - K^{-2}|{}_cF_m|^2) + \Sigma\varepsilon_{ij} + \Sigma\lambda_q G_q.$$

The first term seeks to optimize the agreement between the observed amplitudes ${}_oF_m$, and those calculated from the model ${}_cF_m$; K is a scaling factor which places the observed amplitudes on the same scale as the calculated values, W_m a weight, and the summation is carried out over the m independent reflections. The second term provides a simple quadratic approximation to the non-bonded repulsions and is applied to all pairwise interactions between atoms i and j, when the distance between them is less than some "standard" values (Smith & Arnott, 1978; Guss *et al.*, 1975). The final term includes a set of co-ordinate constraint equations G_q, together with their initially undefined Lagrangian multipliers λ_q, and are used to preserve helix pitch and symmetry as well as, in the case of flexible ring, the ring closure.

Atomic scattering factors for calculating structure factors were calculated using the methods and values given in *International Tables for X-Ray Crystallography* (1974).

(g) *Miscellaneous*

The powder profile refinement program was run on the PRIME 750 computer of the I.N.R.A. Research Centre in Nantes. All the other calculations were performed on the Honeywell–Bull computer of the C.I.C.G. (Centre Interuniversitaire de Calcul de Grenoble). The molecular drawings were realized with the program PITMOS (Pérez & Scaringe, 1986).

3. Structure Determination

(a) *Unit cell content and model building*

As described earlier (Imberty *et al.*, 1987), A-amylose crystals occur in the form of micrometre-sized aggregates of elongated platelets organized in a rosette-like fashion (Fig. 2). Each platelet is a single crystal that yields an electron diffraction diagram (insert in Fig. 2). When probed by X-ray diffraction

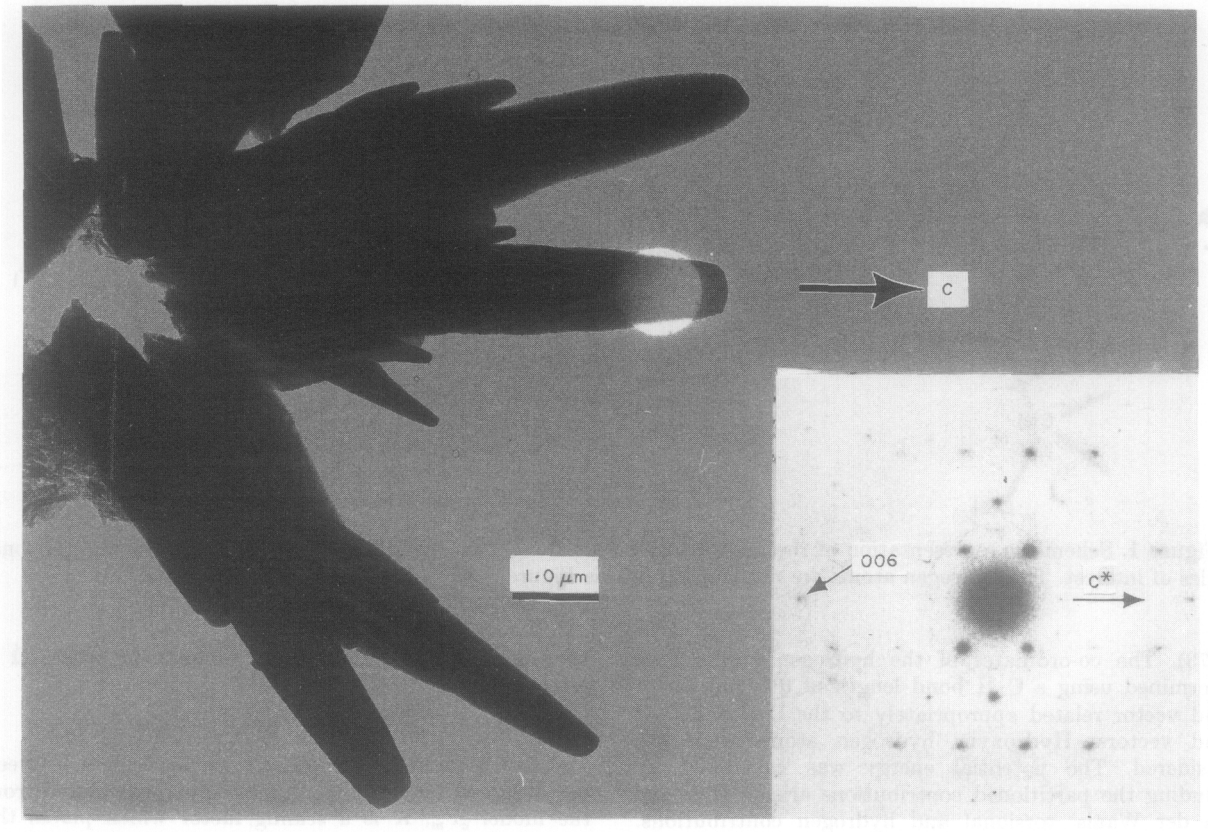


Figure 2. Crystals of A-amylose observed by low dose electron microscopy. Inset: the electron diffraction diagram, recorded under frozen wet conditions corresponds to the circled area in the main Figure. The diagram was obtained with a goniometer setting of 0°. It indexes along the −1 3 0 zone axis in the monoclinic symmetry with the unique *axis* c^* horizontal.

analysis, the crystals, as well as the more abundant polycrystalline powder specimen, give X-ray diagrams that are similar but of a better resolution than those of corn starch (Fig. 3). The similarities displayed by the three diffractograms indicate, without ambiguity, that we are dealing with the same polymorph. When the powder diagrams are decomposed into individual peaks, they yield a series of diffraction lines (Fig. 4) having *d*-spacings which match closely those of the reflections reported by Wu & Sarko (1978*b*).

All these sets of diffraction data yielded the same orthorhombic-like unit cell (Table 1) having a volume of about 2·2 nm^3. The unit cell parameters derived from the electron diffraction experiments are consistently smaller than those obtained from X-ray diffraction experiments on powder or fibre samples. This difference can be explained by the shrinkage of the unit cell at low temperature (−170°C), which was used to record the electron diffractograms. According to the measured density of the single crystals ($d = 1{\cdot}48$), which agrees with the reported density of the fibre ($d = 1{\cdot}51$), 12 glucose residues together with a few possible water molecules can be accommodated in the unit cell.

The amount of diffraction data is not sufficient to attempt a structure elucidation through conventional crystallographic methods. In such a case, one can only proceed by building structural models by conformational analysis methods and then testing the predictions based on these models against experimental data. For an observed fibre repeat *c* of 1·069 nm, computer modelling on α(1 → 4) linked glucose units indicated that the most extended conformations of the linkage would give a repeat of the amylose oligomer with $DP = 3$ (maltotriose), while a shallower one could give a maltohexaose ($DP = 6$). On the assumption that homopolymers have a regular helical conformation

Table 1

Comparison of cell dimensions obtained by different diffraction methods

	Orthorhombic cell			Monoclinic cell
	Fibre X-ray	Crystals electrons	Powder X-ray	Powder X-ray
a(nm)	1·190	1·116	1·172	2·125
b(nm)	1·770	1·751	1·772	1·172
c(nm)	1·052	1·045	1·069	1·069
α(°)	90	90	90	90
β(°)	90	90	90	90
γ(°)	90	90	90	123·48
$V(nm^3)$	2·216	2·042	2·218	2·218
T(°C)	20	−170	20	20

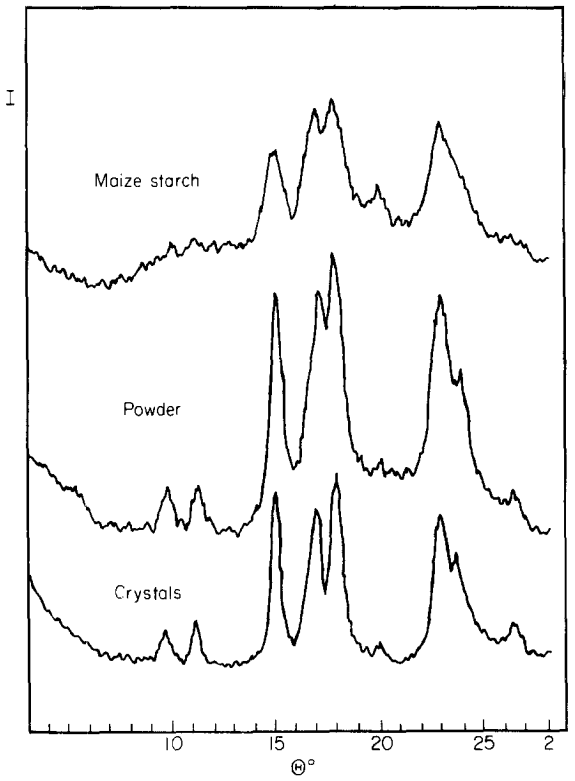


Figure 3. Wide-angle X-ray powder diffraction patterns recorded on: native corn starch; powder obtained by recrystallization from a solution of amylosic fragments of $DP = 15$; powder of single crystals.

(Natta & Corradini, 1959), three possibilities have to be tested against their packing constraints.

(1) Four single 3-fold helices ($n = 3$, $h = 0{\cdot}355$ nm);

(2) two 6-fold helices ($n = 6$, $h = 0{\cdot}178$ nm);

(3) two double 6-fold helices repeating in $2c = 2{\cdot}138$ nm. In order to restore the observed repeat of 1·069 nm, each double helix has to be parallel-stranded, the single strands ($n = 6$, $h = 0{\cdot}355$ nm) being related through a 2-fold axis coincident with the helix axis.

The first hypothesis can be readily discarded through computation of intrachain energy. As seen in Figure 5, the intersect of iso-$h = 0{\cdot}355$ nm with iso-$n = 3$ lies outside the low energy region of the map. As for the second case, it is impossible to pack 6-fold helices into the unit cell because of numerous short interchain contacts which cannot be relieved. The only remaining hypothesis is that of the double parallel-stranded helix, either left or right-handed. Previous attempts to elucidate this structure (Wu & Sarko, 1978*b*) focused on the right-handed situation. Data on other polysaccharides (Pérez & Vergelati, 1985) would tend to favour the occurrence of a left-handed chirality, such as the one proposed by French & Murphy (1977). As shown on Figure 4, the conformation of the intersection of $n = 6$ with $h = -0{\cdot}355$ nm (left-handed chirality) has a lower intramolecular energy than its right-handed counterpart ($n = 6$ and

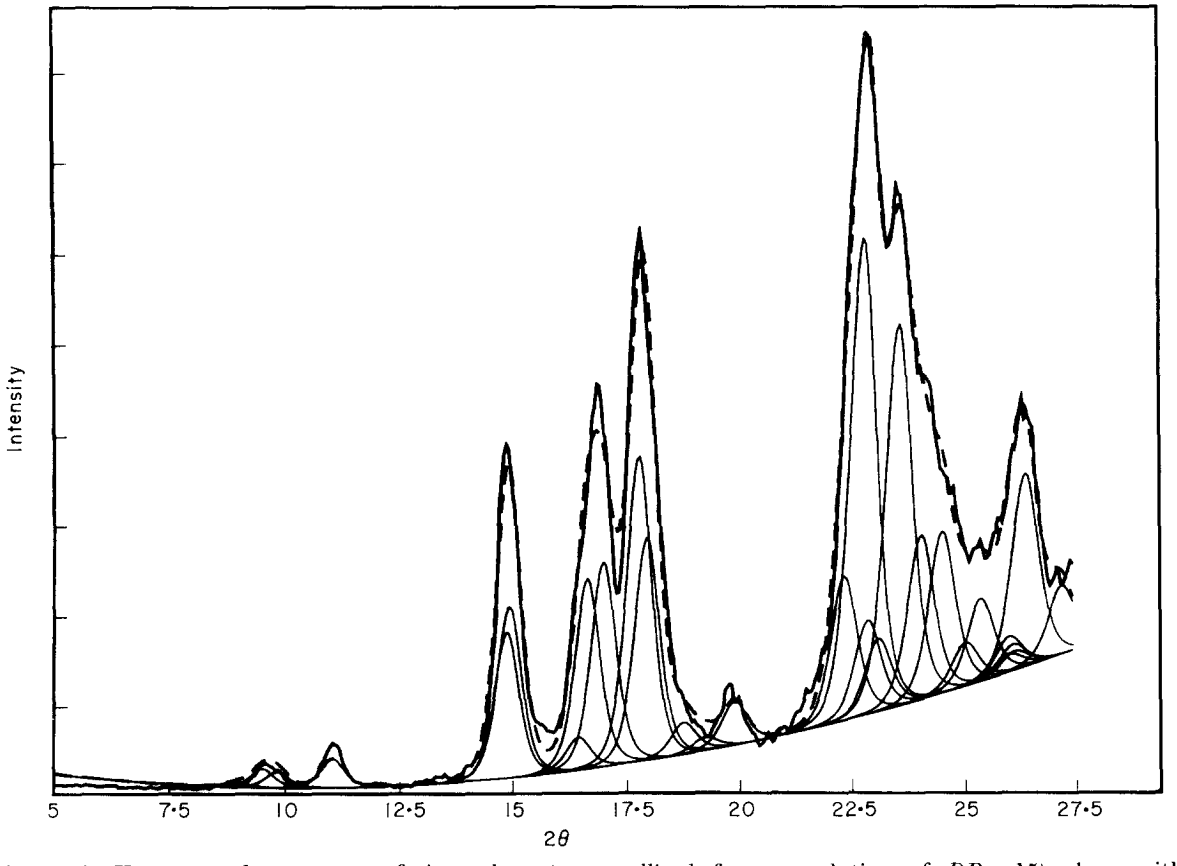


Figure 4. X-ray powder pattern of A-amylose (recrystallized from a solution of $DP = 15$) along with its decomposition into individual diffraction peaks. The observed (————) pattern and the one corresponding to the summation of the calculated (– – – –) peak envelopes are indicated.

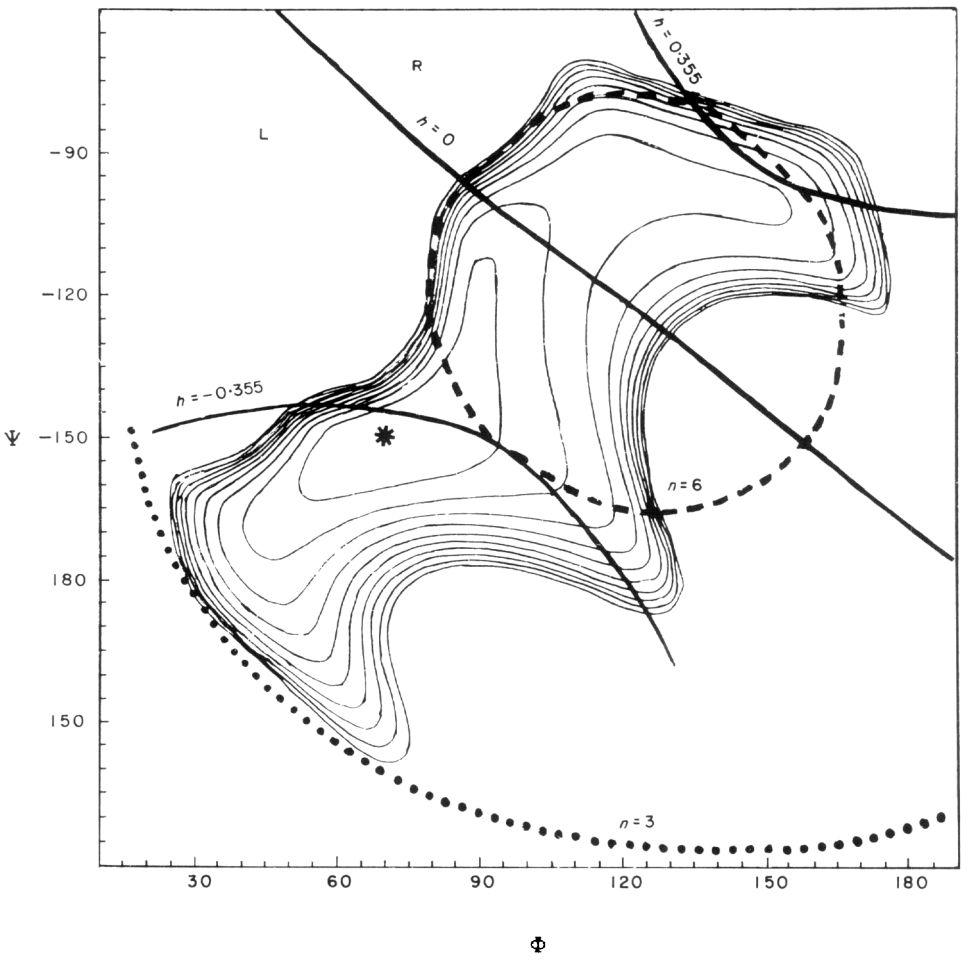


Figure 5. Selected iso-n and iso-h contours for a maltose unit, superimposed on the low energy region of the classical iso-energy map of the $\alpha(1 \rightarrow 4)$ linkage ($\tau = 117{\cdot}5°$). The 2-dimensional iso-energy map was calculated using 5 deg. increments for Φ and Ψ. With respect to the relative energy minimum (*), the iso-energy contours are drawn by interpolation of 1 kcal/mol. The iso-$h = 0$ contour divides the map into 2 regions corresponding to right-handed (R) and left-handed (L) chirality.

$h = +0{\cdot}355$ nm). In fact, the conformation of the glycosidic linkage corresponding to the intersection of $n = 6$ and $h = -0{\cdot}355$ nm, corresponds to one of the stable energy minima calculated for maltose ($DP = 2$) (Pérez & Vergelati, 1987). It is also reminiscent of the observed conformation of a crystalline linear amylose oligomer (Pangborn *et al.*, 1985). Assembling two such parallel single strands through a 2-fold symmetry operation permits formation of a double helix having no short contacts. Thus, on the grounds of energy computation, the occurrence of a left-handed chirality seems to be more likely. Nevertheless, four possible structural models, corresponding to either right or left-handed chirality with the primary hydroxyl groups being set in a *gauche–gauche* (GG)† or a *gauche–trans* (GT) orientation have to be tested against packing and diffraction constraints.

† Abbreviations used: GG, *gauche–gauche*; GT, *gauche–trans*.

(b) *Space group determination and packing analysis*

Electron diffraction data provided the symmetry elements relating two double helices in the unit cell. Informative patterns were obtained by tilting the crystals either along or perpendicular to the platelets' long axes. As described earlier (Imberty *et al.*, 1987), the c axis of the unit cell is parallel to the long axis of the crystals. A rotation about such an axis by + or $-60°$ allows the recording of a series of tilted diagrams which always display an orthogonal symmetry and have in common the c^* axis. Some of these diagrams are shown in Figures 2 (inset) and 6; in Figure 2 is displayed the pattern occurring frequently when no tilt is applied, whereas in Figure 6, diagrams corresponding to $-36°$, $-10°$ and $+20°$ are presented. The remarkable feature of these diagrams, as well as a series of others that are not represented here, is that only reflections with indices $h+k+l = 2n$ are observed. This differs substantially from the data of Wu & Sarko (1978*b*) and proves that some of their

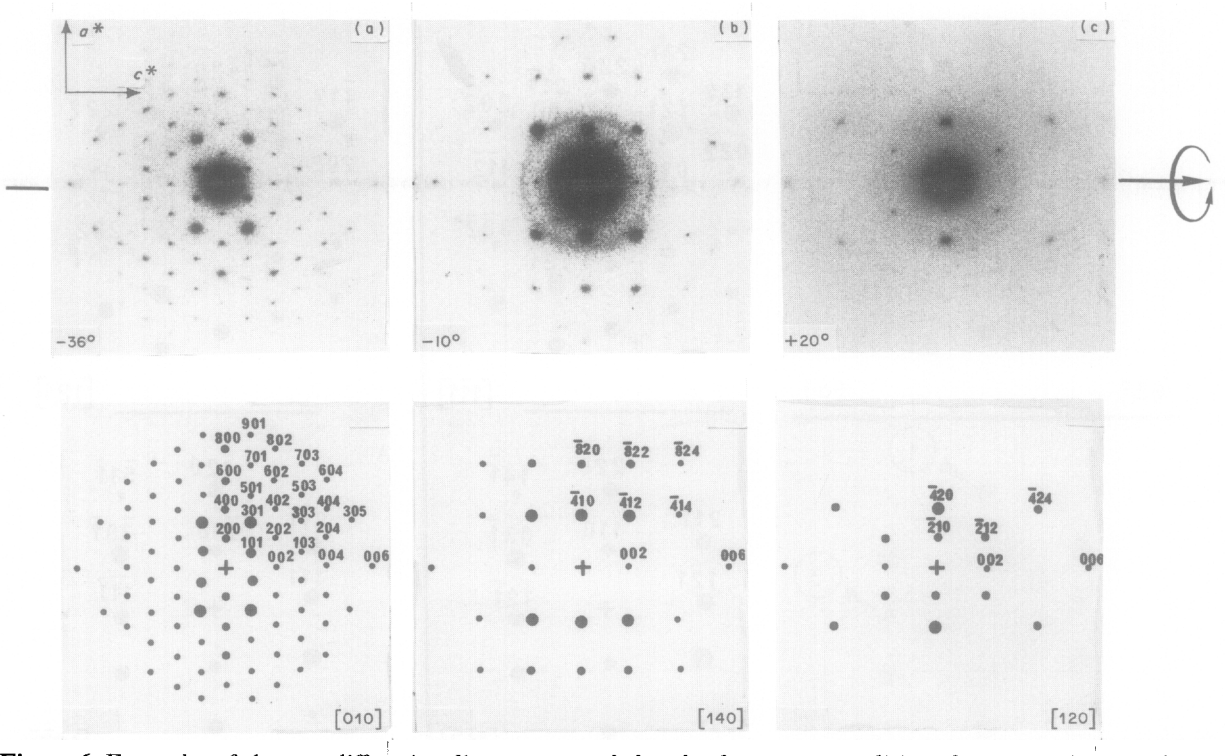


Figure 6. Examples of electron diffraction diagrams recorded under frozen wet conditions from crystals rotated about their long axis. In (a) the microscope goniometer setting was −36°, in (b) −10° and in (c) +20°. The indexations and the corresponding zone axes of these diagrams are also presented.

guesses in fibre diagram indexation were incorrect. In particular, our data indicate, without any ambiguity, that their 0 3 0 (strong) and 2 0 1 (strong) should be indexed as 2 0 0 (strong) and 0 3 1 (strong).

The systematic absences derived from the electron diffractograms are indicative of a body centred lattice, which in the orthorhombic case for chiral systems must be either $I222$ or $I2_12_12_1$. These two space groups, however, are of high symmetry and none of them is able to accommodate two double-helical chiral polymeric chains within the unit cell. For this reason, one must select a system of lower symmetry which is compatible with the observed systematic absences. The only solution remaining is the face-centred monoclinic space group $B2$, with the fibre axis c as the unique monoclinic axis. The unit cell has to undergo an adequate transformation of a and b as presented in Figure 7. In this new system, the unit cell parameters become $a = 2{\cdot}124$ nm, $b = 1{\cdot}172$ nm, $c = 1{\cdot}069$ nm and $\gamma = 123{\cdot}5°$. After re-indexation of the various electron diffraction diagrams, all observed reflections satisfy the imposed conditions for the $B2$ space group.

The confirmation of the monoclinic symmetry was given by recording electron diffractograms on crystals that were tilted perpendicular to the platelets' long axes. Typically, if one had an orthorhombic symmetry, reflections with indices hkl should have the same intensities as those with indices $\bar{h}$-kl. In such a situation, by tilting the crystals about the row (110)–(220) by 63° and 33° (Fig. 8), one should record reflections such as −1-2-1 and −1 2-1 of equal intensities. This is not the case as it can be clearly seen in Figure 8(a) the reflection indexed −1-2-1 is roughly twice as intense as the 2 2 0, whereas in 8(b), −1 2-1 is

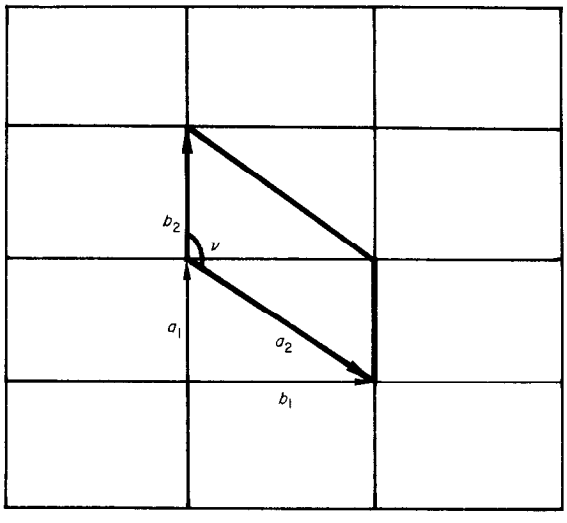


Figure 7. Transformation from orthorhombic (a_1, b_1) to monoclinic (a_2, b_2) unit cell, the c axis remaining unchanged. For a given (h_1, k_1, l_1) reflection in the orthorhombic cell, the corresponding (h_2, k_2, l_2) in the monoclinic cell is obtained using the following matrix operation:

$$\begin{matrix} h_2 \\ k_2 \\ l_2 \end{matrix} = \begin{pmatrix} -1 & -1 & 0 & h_1 \\ 1 & 0 & 0 & k_1 \\ 0 & 0 & 1 & l_1 \end{pmatrix}$$

The condition for systematic absences ($h_1 + k_1 + l_1 = 2n$) in the orthorhombic-like cell becomes in the monoclinic cell: $h_2 + l_2 = 2n$, indicating a face-centred space group.

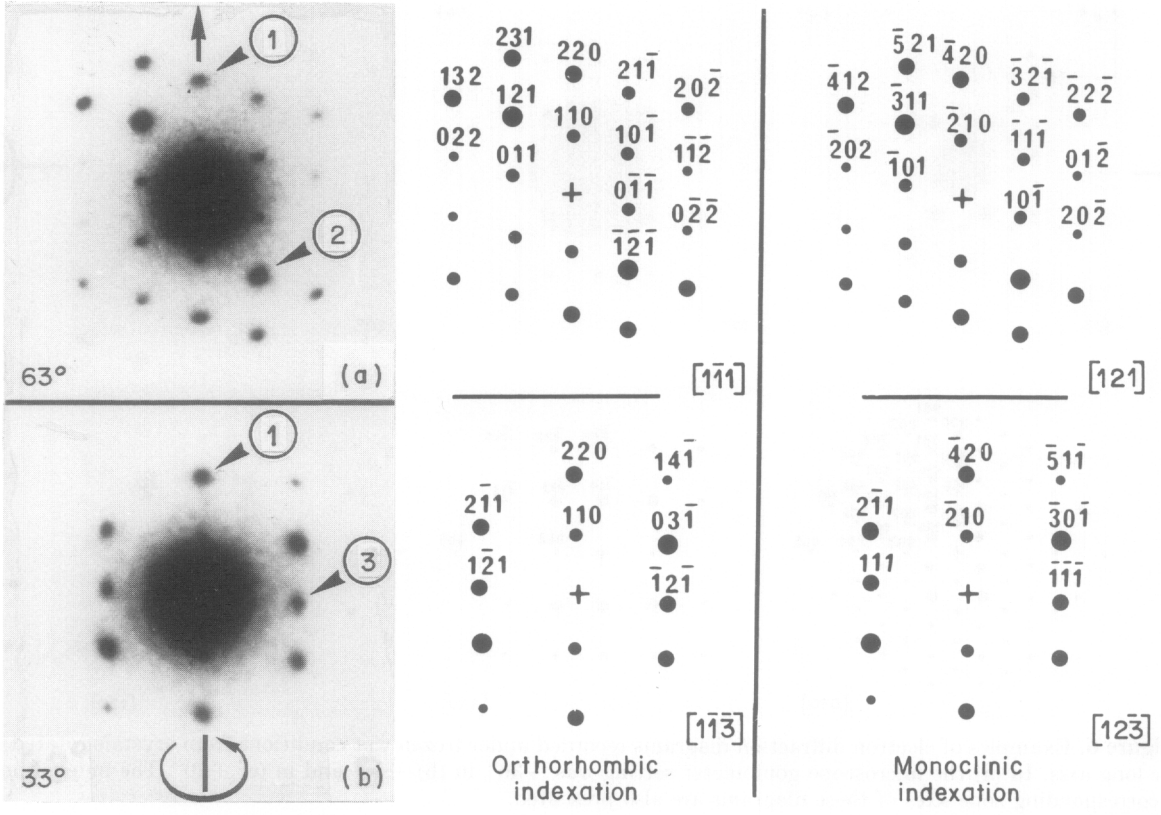


Figure 8. Examples of electron diffraction diagrams recorded under frozen wet conditions from crystals rotated about an axis perpendicular to their long direction. In (a), reflection 2 is roughly twice as intense as that in *1*. In (b), reflection *3* is slightly less intense than *1*. *2* and *3*, which index as −1-2-1 and −1 2-1 in an orthorhombic system, should have equivalent intensities. This is not the case, hence the monoclinic indexation is favoured.

somewhat weaker than the 2 2 0. Therefore, at least in this case, a given F_{hkl} is different from F_{h-kl}. Since, on the other hand, all the diagrams tilted around c^* have orthogonal symmetry, it can be concluded that A-starch is indeed monoclinic, with c as the unique axis.

In the $B2$ space group, the asymmetric unit is made up of a maltotriose moiety. The double helices have to sit on the crystallographic 2-fold axes which are located, respectively, at the corner of the cell and in the middle of the face (a, c). The double helices are related by a simple translation of $a/2+c/2$ and their parallel arrangement is imposed by the symmetry elements of the space group.

(c) *Structure refinement*

(i) *Helices chirality and orientation*

The four putative models (left-handed and right-handed either GG or GT) were checked against both packing constraints and experimental data. Despite the fact that the true crystallographic repeat is a maltotriose unit ($DP = 3$), the first steps of the refinement process were conducted with maintaining a perfect helical conformation for the chain. Furthermore, in the $B2$ space group, only the variable μ for orientation about the unique axis has to be found, since the origin along the c direction is not fixed. Calculations were performed in two independent runs with: (1) diffraction intensities from powder; (2) diffraction intensities from fibre, taken from the work of Wu & Sarko (1978*b*) after appropriate re-indexing of the reflections.

The electron diffraction intensities were not utilized quantitatively in the structure refinement; however, an overall check between these intensities and the observed X-ray diffraction counterparts was performed and a satisfactory agreement was found.

For each model, the two independent runs converged to the same orientation of the helices. In all cases, the number of short contacts between the two strands of the double helix and between the different double helices was calculated. The model with left-handed double helices having a GG orientation of the primary hydroxyl groups had much better three-dimensional R-factor values (from both powder and fibre diffraction data) than the other ones (Table 2). Based on these considerations, as well as on the analysis of the intra-molecular and intermolecular energy, the left-handed double-helical structure having a *gauche-gauche* conformation of the primary hydroxyl groups (Fig. 9) was the only one to be considered for further refinement.

(ii) *Location of water*

Calculated densities of this model are 1·46, 1·52 and 1·58, respectively, with 0, 4 and 8 water

Table 2

Comparison of different models of A-amylose, for their 3-dimensional R*-factors, and their packing features before final refinement*

	Right-handed helix		Left-handed helix	
	GG	GT	GG	GT
3D† *R*-factors (powder)	0·42	0·38	0·34	0·40
3D *R*-factors (fibre)	0·40	0·40	0·26	0·37
Interstrand packing:				
Number of contacts	0	0	10	11
Energy	0	0	1·77	1·21
Inter-helices packing:				
Number of contacts	112	96	35	27
Energy	73·9	41·2	17·1	11·0

Energy is given in arbitrary units.
† 3-dimensional.

molecules per unit cell. Judging by the measured density of the crystals ($d = 1{\cdot}48$) and the reported density of fibre ($d = 1{\cdot}51$) (Wu & Sarko, 1978*b*), four water molecules can be accommodated per unit cell, i.e. one per asymmetric unit. The projection of this molecule in the equatorial plane was determined by calculating the two-dimensional *R*-values during a step displacement of this molecule in the asymmetric unit. The water molecules are localized between the amylosic chains were there is sufficient place to accommodate them. The position of the water molecule along the *c* axis could not be determined clearly through X-ray refinement, perhaps because of a possible delocalization along the column between the chains. The final location was assessed through contact optimization.

(iii) *Final refinement*

In the last step of this study, the helical constraint on the maltotrioside repeating unit was relieved. The valence and torsion bond angles at the glycosidic bridges and the exo-cyclic torsion angles of the primary hydroxyl groups were allowed to vary in a stiffly elastic fashion. By this process, all attractive interactions were improved without altering the atomic positions or the X-ray agreement in any major fashion. The final structure (for the fractional atomic co-ordinates of the asymmetric unit cell content see Table 3) is characterized by a very good packing energy with no short contact (the shortest one being of 0·192 nm between 2 hydrogen atoms of different double helices). The final 3-dimensional crystallographic factors are $R = 0{\cdot}27$ and $R = 0{\cdot}24$ for powder and fibre diffraction intensities, respectively (in 2-dimension, using *hk*0 reflections only, they are, respectively, 0·17 and 0·19). The scaled observed and calculated structure amplitudes for the powder and fibre data sets are given in Tables 4 and 5.

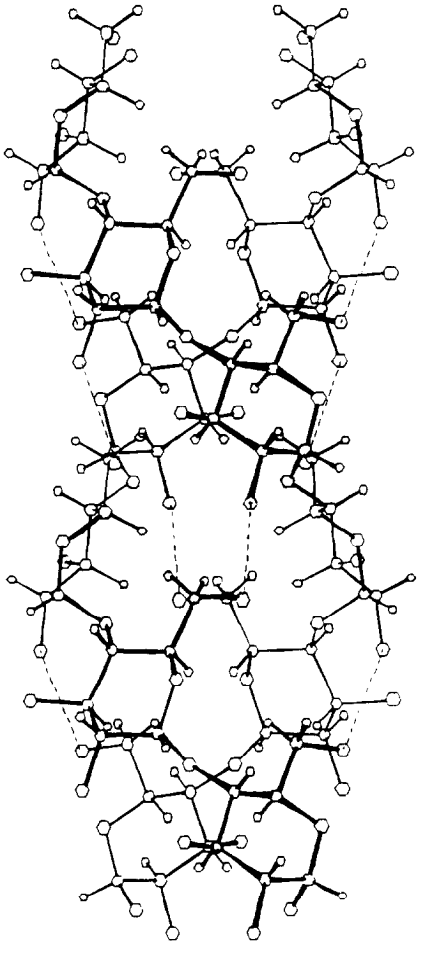


Figure 9. Molecular drawing of the double helix generated by the association of left-handed single amylosic strands: each one being a 6-fold helix repeating in $2c = 2{\cdot}138$ nm. All the primary hydroxyl groups are fixed in a *gauche–gauche* conformation. Hydrogen bonds are represented as broken lines.

4. Discussion

All crystalline samples of A-amylose investigated here, i.e. single crystals, polycrystalline powder and fibres, exhibit the same wide-angle scattering features and therefore the same crystal and molecular organization as the crystalline components of native starch granule. It is now currently accepted that in the starch granule, the branched polymer, amylopectin, is responsible for the diffraction feature, the short chains being associated in crystalline regions, while the branch points are located in amorphous regions (Lineback, 1986). The other starch component, linear amylose, might be incorporated in the amorphous regions. Thus, the structural description which follows is valid for recrystallized substrates, as well as for the native ones.

(a) *Structural analysis*

The present work corroborates previous findings (Wu & Sarko, 1978*b*), in the sense that the true

Table 3

Fractional co-ordinates of the asymmetric unit content

Atom	*x/a*	*y/b*	*z/c*
$C_{(1)}$	0·1577	−0·0704	0·2659
$C_{(2)}$	0·2218	0·0785	0·2694
$C_{(3)}$	0·2049	0·1693	0·1949
$C_{(4)}$	0·1779	0·1088	0·0649
$C_{(5)}$	0·1125	−0·0394	0·0714
$C_{(6)}$	0·0853	−0·1074	−0·0547
$O_{(1)}$	0·0955	−0·0906	0·3341
$O_{(2)}$	0·2430	0·1220	0·3949
$O_{(3)}$	0·2736	0·3004	0·1872
$O_{(4)}$	0·1544	0·1864	0·0000
$O_{(5)}$	0·1361	−0·1164	0·1401
$O_{(6)}$	0·1463	−0·0857	−0·1323
$H_{(1)}$	0·171	−0·132	0·303
$H_{(2)}$	0·265	0·086	0·228
$H_{(3)}$	0·158	0·166	0·241
$H_{(4)}$	0·218	0·111	0·020
$H_{(5)}$	0·069	−0·040	0·115
$H_{(61)}$	0·046	−0·211	−0·046
$H_{(62)}$	0·066	−0·061	−0·096
$C_{(1')}$	−0·0445	−0·3668	0·6086
$C_{(2')}$	0·0359	−0·3287	0·6049
$C_{(3')}$	0·0848	−0·1986	0·5300
$C_{(4')}$	0·0509	−0·2169	0·4004
$C_{(5')}$	−0·0287	−0·2514	0·4141
$C_{(6')}$	−0·0693	−0·2750	0·2914
$O_{(1')}$	−0·0490	−0·2696	0·6788
$O_{(2')}$	0·0629	−0·3203	0·7281
$O_{(3')}$	0·1555	−0·1846	0·5220
$O_{(5')}$	−0·0729	−0·3772	0·4852
$O_{(6')}$	−0·0783	−0·3889	0·2236
$H_{(1')}$	−0·080	−0·456	0·648
$H_{(2')}$	0·035	−0·403	0·562
$H_{(3')}$	0·086	−0·114	0·574
$H_{(4')}$	0·050	−0·290	0·358
$H_{(5')}$	−0·024	−0·168	0·459
$H_{(61')}$	−0·126	−0·315	0·305
$H_{(62')}$	−0·044	−0·188	0·250
$C_{(1'')}$	−0·2090	−0·3072	0·9301
$C_{(2'')}$	−0·1890	−0·4122	0·9387
$C_{(3'')}$	−0·1192	−0·3826	0·8640
$C_{(4'')}$	−0·1206	−0·3244	0·7376
$C_{(5'')}$	−0·1389	−0·2166	0·7392
$C_{(6'')}$	−0·1474	−0·1730	0·6109
$O_{(1'')}$	−0·1571	−0·1897	0·9996
$O_{(2'')}$	−0·1886	−0·4474	1·0654
$O_{(3'')}$	−0·1179	−0·5017	0·8481
$O_{(5'')}$	−0·2103	−0·2715	0·8030
$O_{(6'')}$	−0·2029	−0·2869	0·5366
$H_{(1'')}$	−0·259	−0·339	0·963
$H_{(2'')}$	−0·229	−0·494	0·896
$H_{(3'')}$	−0·072	−0·299	0·917
$H_{(4'')}$	−0·161	−0·396	0·688
$H_{(5'')}$	−0·094	−0·132	0·784
$H_{(61'')}$	−0·164	−0·105	0·616
$H_{(62'')}$	−0·099	−0·134	0·571
W	0·1673	0·6881	0·1928

Table 4

Observed and calculated structure amplitudes from powder diffraction data for the refined model of A-amylose

hkl	F_{obs}	F_{calc}	*hkl*	F_{obs}	F_{calc}
0 1 0			−2 0 2	73	13
−2 1 0	47	55			
			−4 0 0	133	198
−1 0 1	87	133			
			−2 2 2	241	203
−2 0 0	81	100			
			1 2 1		
−1 1 1	110	169	−5 2 1	439	279
1 1 1			3 1 1		
−3 1 1	243	269	−5 1 1	194	289
−2 2 0	264	283	−2 3 0		
			−4 3 0	171	100
2 1 0					
−4 1 0	277	283	2 1 2		
			−4 1 2	392	277
−3 0 1	288	328			
			−3 3 1	257	178
−1 2 1					
−3 2 1	352	362	0 2 2		
			−4 2 2	255	132
0 2 0					
−4 2 0	300	365	2 2 0		
			−6 2 0	133	209
0 1 2					
−2 1 2	114	86			

single-helical repeat occurs at 2·138 nm for six glucose residues. Due to the double-helical nature of the duplex, the asymmetric unit content is a maltotriose repeat, each glucose being in a $^4C_1$ chair conformation. The glycosidic (Φ, Ψ) torsional angles, the valence angles τ and the orientations of the primary hydroxyl groups are given in Table 6. In all residues, the conformation about $O_{(6)}$ is *gauche–gauche*, in agreement with the general rules derived from a large survey of the occurrence of stable conformers in *gluco*pyranoses (Marchessault & Pérez, 1979). The torsional and valence angles (Table 6) exhibit slightly different conformations resulting from small deviations away from perfect helical symmetry. They all displayed a left-handed chirality. Nevertheless, these three conformations correspond to stable relative orientations of contiguous $\alpha(1 \rightarrow 4)$ linked glucose residues: the magnitude of the calculated energy per disaccharide is, at most, 0·6 kcal/mol higher than the one associated with the lowest conformer (Pérez & Vergelati, 1987). Such conformations result in non-bonded distances of 0·329 nm, 0·345 nm and 0·337 nm between the atoms $O_{(2)}$ of one residue and $O_{(3)}$ of the next one for the three linkages, which is too long for intramolecular hydrogen bonding. The distances between the hydrogen atoms $H_{(1)}$ and $H_{(4)}$ about the interglycosidic bridges are reported in Table 6. All the conformational parameters are close to the corresponding ones observed in the crystal structure of methyl-α-maltotriose·$4H_2O$ (Pangborn *et al.*, 1985). The overall chain conformation of each single strand is also not drastically different from the one found for amylose triacetate (Sarko & Marchessault, 1967; Brisse *et al.*, 1982).

The crystal structure of A-amylose is stabilized by an extensive system of hydrogen bonds described in Table 7. Our inability to locate the hydrogen atoms bonded to either the hydroxylic oxygen atoms or to the water oxygen atom leaves room for ambiguity and does not allow a

Table 5

Observed and calculated structure amplitudes from fibre diffraction data for the refined model of A-amylose

hkl†	*hkl*‡	F_{obs}	F_{calc}	*hkl*†	*hkl*‡			*hkl*†	*hkl*‡		
−2 1 0				−8 1 0				−2 0 2	0 2 2	25	12
0 1 0	1 1 0	11	55	6 1 0	1 7 0						
				−8 4 0				−2 2 2	2 0 2		
−2 0 0	0 2 0	62	99	0 4 0	4 4 0	154	155	−4 1 2			
								2 1 2	1 3 2		
−2 2 0	2 0 0	307	280	−1 0 1	0 1 1	60	132	−4 2 2			
								0 2 2	2 2 2	391	355
−4 1 0				−1 1 1	1 0 1	106	168				
2 1 0	1 3 0	334	259					−4 0 2	0 4 2	137	30
				−3 1 1							
−4 2 0				1 1 1	1 2 1	253	266	−4 3 2			
0 2 0	2 2 0	419	359					−2 3 2	3 1 2	288	204
				−3 0 1	0 3 1						
−4 0 0	0 4 0	264	194	−3 2 1				−6 2 2			
				−1 2 1	2 1 1	437	478	2 2 2	2 4 2		
−4 3 0								−6 1 2			
−2 3 0	3 1 0	282	97	−5 2 1				4 1 2	1 5 2	271	279
				1 2 1	2 3 1						
−6 2 0				−5 1 1				−1 0 3	0 1 3	23	68
2 2 0	2 4 0	225	203	3 1 1	1 4 1						
				−3 3 1	3 0 1	431	424	−1 1 3	1 0 3	24	110
−6 1 0											
4 1 0	1 5 0			−5 3 1				−3 1 3			
−6 3 0				−1 3 1	3 2 1			1 1 3	1 2 3	64	78
0 3 0	3 3 0	183	170	−5 0 1	0 5 1	208	152				
								−3 2 3			
−4 0 0	4 0 0			−7 2 1				1 1 3	2 1 3		
−6 6 0	0 6 0	308	170	3 2 1	2 5 1			−3 0 3	0 3 3	178	192
				−7 3 1							
−6 4 0				1 3 1	3 4 1			−5 2 3			
−2 4 0	4 2 0	183	159	−5 4 1				1 2 3	2 3 3	249	264
				−3 4 1	4 1 1	244	361				
−8 2 0								−5 1 3			
4 2 0	2 6 0			−7 1 1				3 1 3	1 4 3		
−8 3 0				5 1 1	1 6 1	209	198	−3 3 3	3 0 3	263	368
2 3 0	3 5 0	279	257								
				−2 1 2				−5 3 3			
				0 12	1 1 2	50	84	−1 3 3	2 3 3		
								−5 0 3	0 5 3	116	159

† *hkl* in the new monoclinic cell.
‡ *hkl* in the orthorhombic cell before re-indexing.

description of the network in terms of donor and acceptor. Within the double helix, interstrand stabilization is achieved without any steric conflict and through the occurrence of the $O_{(2)} \cdots O_{(6)}$ type of hydrogen bonding. There are two such bonds per glucose residues, i.e. six per maltotriose unit ranging from 0·266 nm to 0·293 nm. However, these hydrogen bonds are not the only force which stabilize the double helix, since computation of the interstrand energy shows that 60% occurs from van der Waals' interactions. The inner dimension of the double helix cavity is 0·35 nm in diameter which does not allow any water molecule to pack into the hole, while the outer diameter is about 1·03 nm.

A projection of the structure onto the (*a*, *b*) plane is shown in Figure 10, in which the pseudo-hexagonal packing of double helices is obvious; each double helix is being surrounded by six neighbours. As can be seen in Figure 11, it is clear that such a three-dimensional arrangement obeys the law of close packing. The *B*2 space group appears to be particularly suited for dense packing of such molecules, since the symmetry operations (especially the *c*/2 translation) permit the "nesting" of neighbouring double helices having pronounced "crests" and "troughs". The cohesion between these double helices is further maintained through hydrogen bonding involving distances ranging from 0·263 nm to 0·278 nm (Table 7). The water molecule, acting as a donor to the ring oxygen atoms $O_{(5)}$ and $O_{(5'')}$, and as an acceptor for the primary hydroxyl group oxygen atom $O_{(6)}$ of three neighbouring double helices, stabilizes the structure through supplementary hydrogen bonds. It must be noted that all the hydrogen bonding possibilities for a given maltotriose unit are not fulfilled, since each of the secondary hydroxyl oxygen atom $O_{(3)}$ is capable of participating in at least one more hydrogen bond. It is, however, surprising to see the limited amount of water present in the structure, as well as its minor role in establishing the three-dimensional features.

Table 6
Conformational features of the repeating unit

A. *Glycosidic bond angles (deg.)*

$C_{(1)} - O_{(1)} - C_{(4')} = 117{\cdot}0$
$C_{(1')} - O_{(1')} - C_{(4'')} = 113{\cdot}6$
$C_{(1'')} - O_{(1'')} - C_{(4)}\dagger = 120{\cdot}5$

B. *Torsion angles (deg.) about the glycosidic linkage*

$O_{(5)} - C_{(1)} - O_{(1)} - C_{(4')} = 91{\cdot}8$
$C_{(1)} - O_{(1)} - C_{(4')} - C_{(5')} = -153{\cdot}2$

$O_{(5')} - C_{(1')} - O_{(1')} - C_{(4'')} = 85{\cdot}7$
$C_{(1')} - O_{(1')} - C_{(4'')} - C_{(5'')} = -145{\cdot}3$

$O_{(5'')} - C_{(1'')} - O_{(1'')} - C_{(4)}\dagger = 91{\cdot}8$
$C_{(1'')} - O_{(1'')} - C_{(4)}\dagger - C_{(5)}\dagger = -151{\cdot}3$

C. *Distances (nm) $H_{(1)}$–$H_{(4)}$ for each glycosidic bond*

$H_{(1)} \cdots H_{(4')} = 0{\cdot}226$
$H_{(1')} \cdots H_{(4'')} = 0{\cdot}223$
$H_{(1'')} \cdots H_{(4)}\dagger = 0{\cdot}239$

D. *Torsion angles (deg.) for the primary hydroxyl groups*

$O_{(5)} - C_{(5)} - C_{(6)} - O_{(6)} = -70{\cdot}7$
$O_{(5')} - C_{(5')} - C_{(6')} - O_{(6')} = -58{\cdot}1$
$O_{(5'')} - C_{(5'')} - C_{(6'')} - O_{(6'')} = -64{\cdot}4$

† Atoms of the following residue of the helix $(-x, -y, z+c)$.

Table 7
Hydrogen bonding in A-amylose

A. *Interstrand hydrogen bonds*

i . . *j*	Symmetry	*d*(nm)
$O_{(2)} \cdots O_{(6'')}$	II	0·293
$O_{(6)} \cdots O_{(2')}$	I−c	0·275
$O_{(2')} \cdots O_{(6)}$	I+c	0·275
$O_{(6')} \cdots O_{(2'')}$	I−c	0·266
$O_{(2'')} \cdots O_{(6')}$	I+c	0·266
$O_{(6'')} \cdots O_{(2)}$	II	0·293

B. *Inter-double-helix hydrogen bonds*

i . . *j*	Symmetry	*d*(nm)
$O_{(2)} \cdots O_{(6)}$	IV	0·263
$O_{(3)} \cdots O_{(3'')}$	III−c	0·278
$O_{(6)} \cdots O_{(2)}$	IV−c	0·263
$O_{(2'')} \cdots O_{(6'')}$	IV−a−b	0·268
$O_{(3'')} \cdots O_{(3)}$	III−a−b	0·278
$O_{(6'')} \cdots O_{(2'')}$	IV−a−b	0·268
$O_{(5)} \cdots W$	I−b	0·277
$O_{(6')} \cdots W$	II	0·294
$O_{(5'')} \cdots W$	III−a−b	0·265

Symmetry code: I, x, y, z; II, $-x, -y, z$; III, $1/2+x, y, 1/2+z$; IV, $1/2-x, -y, 1/2+z$.

(b) *Physicochemical and biochemical considerations*

The double-helical nature of the crystalline component of native starch, already proposed by several authors, is in agreement with its physicochemical properties such as insolubility in cold water and resistance to acid treatment. The spontaneous association in double helices in solution has been postulated to explain a major property of starch: its retrogradation after gelatinization (Matsukura *et al.*, 1983). This behaviour appears to be a ubiquitous phenomenon, governed either by biosynthesis (in the starch granule) or by simple thermodynamic rules (growth of single crystal *in vitro*, retrogradation). The new features of A-amylose structure proposed here are the left-handed chirality, the maltotriose repeat unit, the *gauche–gauche* conformation of the primary hydroxyl group and the parallel arrangement of all the chains in the crystal.

The actual finding that within the double-helical assembly, each strand has a left-handed chirality is not particularly surprising. In fact, the crystalline fibre specimen of A-amylose was obtained by solid-state deacetylation of oriented, crystalline amylose

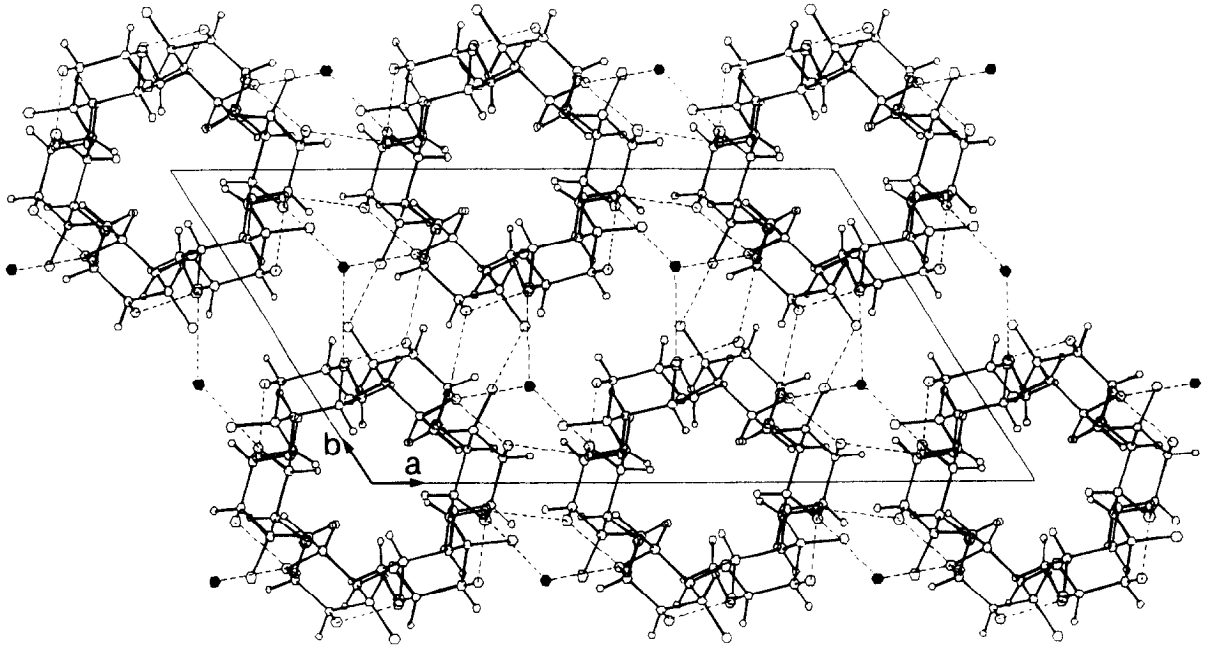


Figure 10. Projection of the structure onto the (*a*, *b*) plane. Hydrogen bonds are indicated as broken lines; (●) indicate water molecules.

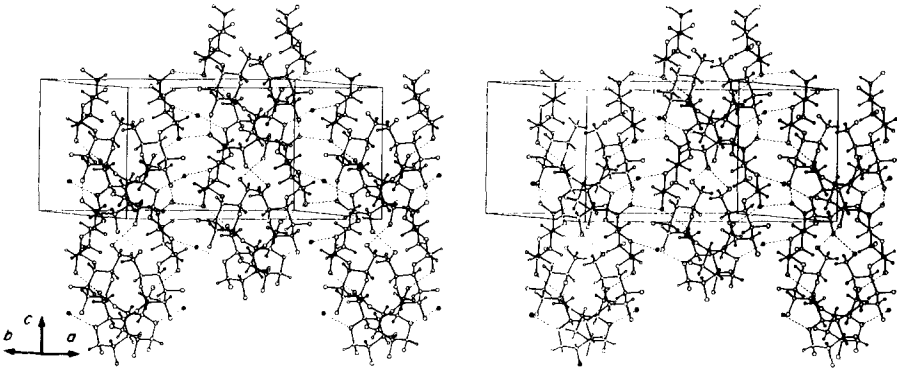


Figure 11. Stereoscopic representation of the 3-dimensional structure of A-amylose. (●) Indicate water molecules; hydrogen bonds are shown as broken lines.

triacetate (ATA) (Wu & Sarko, 1978). ATA has been reported to be a non-integral left-handed 14/11 helix with 4·67 residues per turn (Sarko & Marchessault, 1967). Similarly, a series of alkali–amylose complexes can be obtained during the solid-state deacetylation of ATA (Senti & Witnauer, 1946). For example in the KOH–amylose complex structure (Sarko & Biloski, 1980) the amylosic chain exhibits a left-handed conformation with typical helical parameters $h = -0{\cdot}375$ nm and $n = 6$, corresponding to idealized glycosyl torsional angles $\Phi = 92{\cdot}5°$ and $\Psi = -148{\cdot}9°$ (incidentally, all the crystalline arrangements are based on 3-dimensional packing of single chains). Also, evidence for this type of conformation in aqueous solution has been obtained through high resolution nuclear magnetic resonance (Pérez *et al.*, 1985) and through monochromatic optical rotation experiments on amylose solution (Rees, 1981).

Recently, four independent reports on the high resolution ^{13}C solid-state nuclear magnetic resonance of starch polymorphs have appeared (Gidley & Bociek, 1985; Hewitt *et al.*, 1986; Veregin *et al.*, 1986; Horii *et al.*, 1986). All these works made use of the ^{13}C resonances of the carbon atoms involved in the glycosidic linkage and of $C_{(6)}$ bonded to the primary hydroxyl group. The chemical shifts of these atoms were shown to be particularly sensitive to the molecular conformation and lattice structure. As for the resonances corresponding to the anomeric carbon atoms $C_{(1)}$, a triplet is systematically recorded for A-type polymorph. This is in line with the fact that the asymmetric unit is a maltotriose moiety, which has three different sets of (Φ, Ψ) torsion angles. At this stage it is of course illusory to correlate a particular conformation to a given resonance. The chemical shift of $C_{(6)}$ in A and B-amyloses occurs at about 62·5 (0·5) parts per million. Using short linear and cyclic model compounds, a correlation has been established between chemical shifts and conformation about $C_{(5)}$–$C_{(6)}$ bonds and a *gauche–trans* conformation for both A and B polymorphs has been predicted (Horii *et al.*, 1983). As shown in section (a), above, the occurrence of such conformation is less probable than the *gauche–gauche*, based upon both crystallographic and packing criteria. It may be advocated that the neighbourhood of $C_{(6)}$ in the primary hydroxyl group is strongly influenced by the double-helical structure, and that correlations which were established for low molecular weight compounds may not be valid. In fact, the effect of neighbourhood on charge density and subsequent chemical shift has already been noticed (Henrissat *et al.*, 1987; Inoue *et al.*, 1985).

Several models for amylopectin have been proposed based on the analysis of the products of enzymatic treatment (Duprat *et al.*, 1980; Manners & Matheson, 1981; Hizukuri, 1985). A debranching treatment gives a bimodal distribution of chains regarding their *DP*. The currently accepted models are "tree-like" with two sorts of branches. In A-amylose long branches have an average *DP* of 45 while the short ones (which are dominant) have an average *DP* of 15. These short branches are thought to be associated in double helices and their parallel arrangements represent the crystalline part of the starch granules. In all these "cluster" models, the chains are branched by their reducing end and are therefore all in a parallel register.

(c) *Starch granule structure and biogenesis*

The structure presented here does not represent the whole structure of the starch granule but at least the crystalline part which can be reached by the crystallographic approach. In one of the most recent models proposed for the starch granule structure (Lineback, 1986), the crystalline parts, composed of a parallel arrangement of short branches of amylopectin, are radially arranged in the granule. The amorphous regions are composed of other parts of amylopectin and a long chain of linear amylose.

In starch biosynthesis, the branching of one chain on another is achieved by the Q-enzyme, which has some particular requirements. It acts only on two chains previously associated in a duplex, probably through hydrogen bonding (Borowsky *et al.*, 1975). This behaviour suggests that the branching occurs only after the synthesis of a linear fragment of

amylose, and concomitant association in double helices are achieved.

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